

Numerical Solution of Heat equation (one-dimension)

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \rightarrow \textcircled{1}$$

Suppose we wish to solve eqⁿ $\textcircled{1}$ for $0 < x < a$ and $t > 0$

Under the boundary conditions

$$u(0, t) = \phi(t) \text{ for } t \geq 0$$

$$u(a, t) = \psi(t) \text{ for } t \geq 0$$

where $\phi(t)$ and $\psi(t)$ are specified ~~by~~ functions of t ($t \geq 0$)

and the initial condition

$$u(x, 0) = f(x) \text{ for } 0 < x < a$$

where $f(x)$ is a specified function of x (for $0 < x < a$)

Consider a rectangular mesh in the x - t plane with spacing h along x -direction and k along t direction.

Denoting the mesh point $(x, t) = (ih, jk)$ as simply i, j

we have

$$\frac{\partial u}{\partial t} = \frac{u_{i,j+1} - u_{i,j}}{k} \quad (\text{forward difference})$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2}$$

$\textcircled{1} \Rightarrow$

$$\frac{u_{i,j+1} - u_{i,j}}{k} = \frac{c^2}{h^2} (u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

$$\Rightarrow u_{i,j+1} - u_{i,j} = \frac{kc^2}{h^2} (u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

$$u_{i,j+1} - u_{i,j} = \alpha (u_{i-1,j} - 2u_{i,j} + u_{i+1,j})$$

$$\text{where } \alpha = \frac{kc^2}{h^2}$$

$$u_{i,j+1} = \alpha (u_{i-1,j} + u_{i+1,j}) - 2\alpha u_{i,j} + u_{i,j}$$

$$\Rightarrow \boxed{u_{i,j+1} = \alpha (u_{i-1,j} + u_{i+1,j}) + (1 - 2\alpha) u_{i,j}} \rightarrow (2)$$

This formula enables us to determine the values of u at the $(i, j+1)^{\text{th}}$ mesh point in terms of the known function values at the x_{i-1} , x_i and x_{i+1} at the instant t_j

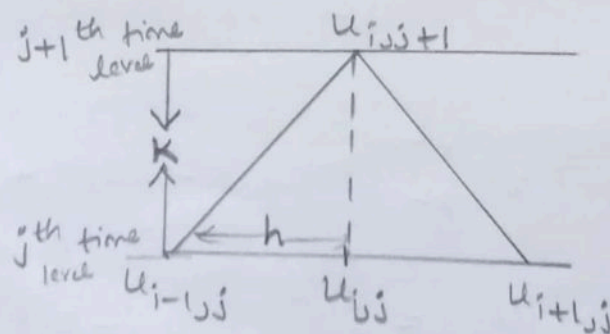
It is a relation between the function values at the two levels $j+1$ and j , therefore called a 2-level formula

Eqⁿ (2) is called Schmidt explicit formula which is valid only for $0 < \alpha \leq \frac{1}{2}$.

Note: when $\alpha = \frac{1}{2}$ in (2)

$$\Rightarrow \boxed{u_{i,j+1} = \frac{1}{2} [u_{i-1,j} + u_{i+1,j}]} \rightarrow (3)$$

Eqⁿ (3) is called Bendre-Schmidt formula.



(i) Consider the heat equation $\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2}$ under the

conditions: (i) $u(0, t) = u(4, t) = 0, t \geq 0$

(ii) $u(x, 0) = x(4-x), 0 < x < 4$

Employ the Bender-Schmidt method with $h=1$ to find the solution of the equation for $0 < t \leq 1$.

Soln: Since $h=1$, the given interval $[0, 4]$, on which

x -varies, is to be divided into 4 equal parts, with $x_0=0$ and $x_4=4$ as boundary points and

$x_i = ih \Rightarrow x_i = i$ for $i=1, 2, 3, \dots$

Now the given boundary conditions $u(0, t) = u(4, t) = 0, t \geq 0$ give

$$u(0, t) = u(x_0, t_j) = u_{0,j} = 0, j=0, 1, 2, 3, \dots$$

$$\Rightarrow u_{0,0} = u_{0,1} = u_{0,2} = u_{0,3} = \dots = 0$$

$$\text{Similarly } u(4, t) = u(x_4, t_j) = u_{4,j} = 0 \text{ for } j=0, 1, 2, 3, \dots$$

$$\Rightarrow u_{4,0} = u_{4,1} = u_{4,2} = u_{4,3} = \dots = 0$$

The given initial condition $u(x, 0) = x(4-x), 0 < x < 4$

$$u(x, 0) = x(4-x)$$

$$u(x_i, 0) = x_i(4-x_i)$$

$$u_{i,0} = ih(4-ih) = i(4-i) \quad (\because h=1) \\ \text{for } i=1, 2, 3, \dots$$

$$u_{1,0} = 1(4-1) = 3, \quad u_{2,0} = 2(4-2) = 4, \quad u_{3,0} = 3(4-3) = 3$$

We note that $c^2 = 2$, $h = 1$, $\alpha = \frac{1}{2}$ ($\because$ Bendre-Schmidt method)

$$\therefore \alpha = \frac{Kc^2}{h^2} \Rightarrow \frac{1}{2} = \frac{2K}{1}$$

$$\Rightarrow \boxed{K = \frac{1}{4}}$$

$\therefore$ By Bendre Schmidt formula, we have

$$u_{i,j+1} = \frac{1}{2} [u_{i-1,j} + u_{i+1,j}] \rightarrow \textcircled{1}$$

For the first time level, we put $j=0$ in $\textcircled{1}$

$$\therefore u_{i,1} = \frac{1}{2} [u_{i-1,0} + u_{i+1,0}]$$

For $i=1, 2, 3$ we get

$$u_{1,1} = \frac{1}{2} [u_{0,0} + u_{2,0}] = \frac{1}{2} [0 + 4] = 2$$

$$u_{2,1} = \frac{1}{2} [u_{1,0} + u_{3,0}] = \frac{1}{2} [3 + 3] = 3$$

$$u_{3,1} = \frac{1}{2} [u_{2,0} + u_{4,0}] = \frac{1}{2} [4 + 0] = 2$$

These expression constitute the solution at first time level $t_1 = \frac{1}{4}$.

For the second time level, we put $j=1$ in $\textcircled{1}$

$$\therefore u_{i,2} = \frac{1}{2} [u_{i-1,1} + u_{i+1,1}]$$

For $i=1, 2, 3$, we get

$$u_{1,2} = \frac{1}{2} [u_{0,1} + u_{2,1}] = \frac{1}{2} [0 + 3] = \frac{3}{2}$$

$$u_{2,2} = \frac{1}{2} [u_{1,1} + u_{3,1}] = \frac{1}{2} [2 + 2] = 2$$

$$u_{3,2} = \frac{1}{2} [u_{2,1} + u_{4,1}] = \frac{1}{2} [3 + 0] = \frac{3}{2}$$

These constitute the solution at the second time level $t_2 = \frac{2}{4}$

For the third time level, we put $j=2$ in ①

$$\therefore u_{i,3} = \frac{1}{2} [u_{i-1,2} + u_{i+1,2}]$$

Put $i=1, 2, 3$

$$u_{1,3} = \frac{1}{2} [u_{0,2} + u_{2,2}] = \frac{1}{2} [0 + 2] = 1$$

$$u_{2,3} = \frac{1}{2} [u_{1,2} + u_{3,2}] = \frac{1}{2} [\frac{3}{2} + \frac{3}{2}] = \frac{3}{2}$$

$$u_{3,3} = \frac{1}{2} [u_{2,2} + u_{4,2}] = \frac{1}{2} [0 + 2] = 1$$

These constitute the solution at third time level $t_3 = \frac{3}{4}$

For the Fourth time level, we put $j=3$ in ①

$$\therefore u_{i,4} = \frac{1}{2} [u_{i-1,3} + u_{i+1,3}]$$

Put $i=1, 2, 3$

$$u_{1,4} = \frac{1}{2} [u_{0,3} + u_{2,3}] = \frac{1}{2} [0 + \frac{3}{2}] = \frac{3}{4}$$

$$u_{2,4} = \frac{1}{2} [u_{1,3} + u_{3,3}] = \frac{1}{2} [1 + 1] = 1$$

$$u_{3,4} = \frac{1}{2} [u_{2,3} + u_{4,3}] = \frac{1}{2} [\frac{3}{2} + 0] = \frac{3}{4}$$

These constitute the solution at the fourth time level $t_4 = \frac{4}{4} = 1$

4th time level	0	$\frac{3}{4}$	1	$\frac{3}{4}$	0
3rd time level	0	1	$\frac{3}{2}$	1	0
2nd time level	0	$\frac{3}{2}$	2	$\frac{3}{2}$	0
1st time level	0	2	3	2	0
	0	3	4	3	0

2) Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ Subject to the conditions

$$u(x, 0) = \sin \pi x, \quad 0 \leq x \leq 1,$$

$u(0, t) = u(1, t) = 0$. Carry out computations for two levels taking $h = \frac{1}{3}$ and $k = \frac{1}{26}$.

Soln: Here $c^2 = 1$, $h = \frac{1}{3}$, $k = \frac{1}{26}$

$$\therefore \alpha = \frac{kc^2}{h^2} = \frac{1}{26} \times 9 = \frac{1}{4}$$

$$\boxed{\alpha = \frac{1}{4}}$$

Since $h = \frac{1}{3}$, the given interval $[0, 1]$, $\begin{array}{cccc} x_0 & x_1 & x_2 & x_3 \\ 0 & \frac{1}{3} & \frac{2}{3} & 1 \end{array}$
on which x -varies is to be divided into 3 equal parts,
with $x_0 = 0$, and $x_3 = 1$ as boundary points.

$$x_i = ih \Rightarrow x_i = \frac{i}{3} \text{ for } i = 1, 2, 3, \dots$$

Now the given boundary conditions

$$u(0, t) = 0$$

$$\Rightarrow u(x_0, t_j) = 0$$

$$u_{0,j} = 0 \text{ for } j = 0, 1, 2, 3$$

$$u_{0,0} = u_{0,1} = u_{0,2} = u_{0,3} = \dots = 0$$

$$u(1, t) = 0$$

$$\Rightarrow u(x_3, t_j) = 0$$

$$u_{3,j} = 0 \text{ for } j = 0, 1, 2, 3$$

$$u_{3,0} = u_{3,1} = u_{3,2} = \dots = 0$$

The given initial condition is $u(x, 0) = \sin \pi x$

$$u(x, 0) = \sin \pi x$$

$$\Rightarrow u(x_i, 0) = \sin \pi x_i$$

$$u_{i,0} = \sin \pi(ih) = \sin\left(\frac{\pi i}{3}\right) \quad \text{for } i=1, 2, 3$$

$$u_{1,0} = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} \quad (\text{use radians})$$

$$u_{2,0} = \sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$u_{3,0} = \sin(\pi) = 0$$

By Schmidt's formula, we have

$$u_{i,j+1} = \alpha [u_{i+1,j} + u_{i-1,j}] + (1-2\alpha) u_{i,j}$$

$$u_{i,j+1} = \frac{1}{4} [u_{i+1,j} + u_{i-1,j}] + \left(1 - \frac{1}{2}\right) u_{i,j}$$

Taking LCM

$$= \frac{1}{4} [u_{i+1,j} + u_{i-1,j} + 2u_{i,j}] \rightarrow \textcircled{1}$$

For the first time level, we put $j=0$ in $\textcircled{1}$

$$u_{i,1} = \frac{1}{4} [u_{i+1,0} + u_{i-1,0} + 2u_{i,0}]$$

For $i=1, 2$

$$u_{1,1} = \frac{1}{4} [u_{2,0} + u_{0,0} + 2u_{1,0}]$$

$$= \frac{1}{4} \left[\frac{\sqrt{3}}{2} + 0 + 2\left(\frac{\sqrt{3}}{2}\right) \right]$$

$$u_{1,1} = 0.6495$$

$$u_{2,1} = \frac{1}{4} [u_{3,0} + u_{1,0} + 2u_{2,0}] = \frac{1}{4} \left[0 + \frac{\sqrt{3}}{2} + 2\left(\frac{\sqrt{3}}{2}\right) \right]$$

$$u_{2,1} = 0.6495$$

For the second time level, put $j=1$ in (1)

$$u_{i,2} = \frac{1}{4} [u_{i+1,1} + u_{i-1,1} + 2u_{i,1}]$$

For $i=1, 2$ we get

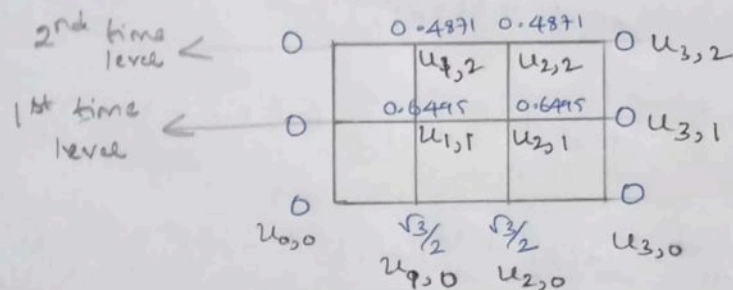
$$u_{1,2} = \frac{1}{4} [u_{2,1} + u_{0,1} + 2u_{1,1}] = \frac{1}{4} [0.6495 + 0 + 2(0.6495)]$$

$$= 0.4871$$

$$u_{2,2} = \frac{1}{4} [u_{3,1} + u_{1,1} + 2u_{2,1}] = \frac{1}{4} [0 + 0.6495 + 2(0.6495)]$$

$$= 0.4871$$

These constitute the solution at the second time level



3) Use Schmidt method to solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$

under the conditions (i) $u(0,t) = u(1,t) = 0, t \geq 0$

(ii) $u(x,0) = \sin \pi x, 0 < x < 1$

by taking $h = \frac{1}{4}$ and $\Delta = \frac{1}{6}$. Carry out three steps in the time level.

Soln: we note that $C^2 = 1, h = \frac{1}{4}, \Delta = \frac{1}{6}$

$$\Rightarrow \Delta = \frac{K C^2}{h^2} \Rightarrow \frac{1}{6} = \frac{K}{(\frac{1}{4})^2}$$

$$\Rightarrow K = \frac{1}{96}$$

Since $h = \frac{1}{4}$, the given interval $[0, 1]$, on which x -varies is to be divided into 4 equal parts, with 1 to 4

$$x_i = ih = \frac{i}{4}, \quad i = 1, 2, 3,$$

x_0	x_1	x_2	x_3	x_4
0	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	1

$$u(0, t) = 0$$

$$\Rightarrow \cancel{u(x_0, t_j)} = 0 \Rightarrow u_{0,j} = 0 \quad \text{for } j = 0, 1, 2,$$

$$u_{0,0} = u_{0,1} = u_{0,2} = u_{0,3} = \dots = 0$$

$$u(1, t) = 0$$

$$\Rightarrow u(x_4, t_j) = 0 \Rightarrow u_{4,j} = 0 \quad \text{for } j = 0, 1, 2, \dots$$

$$u_{4,0} = u_{4,1} = u_{4,2} = u_{4,3} = u_{4,4} = \dots = 0$$

The given initial condition $u(x, 0) = \sin \pi x$, $0 < x < 1$

$$u(x_i, 0) = \sin \pi x_i$$

$$u_{i,0} = \sin \pi(ih) = \sin\left(\frac{\pi i}{4}\right)$$

$$u_{i,0} = \cancel{\sin\left(\frac{\pi i}{4}\right)} \sin\left(\frac{\pi i}{4}\right) \quad \text{for } i = 1, 2, 3$$

$$u_{1,0} = \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$u_{2,0} = \sin\left(\frac{\pi}{2}\right) = 1$$

$$u_{3,0} = \sin\left(\frac{3\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

We require to compute the solution upto 3rd level in time

i.e upto $t = 3k = \frac{1}{32}$

By Schmidt method, we have

$$u_{i,j+1} = \alpha(u_{i+1,j} + u_{i-1,j}) + (1-2\alpha)u_{i,j}$$

$$u_{i,j+1} = \frac{1}{6} (u_{i+1,j} + u_{i-1,j}) + (1 - \frac{1}{2}) u_{i,j}$$

$$u_{i,j+1} = \frac{1}{6} [u_{i+1,j} + u_{i-1,j} + 4u_{i,j}] \rightarrow (1)$$

For the first time level $t=k=\frac{1}{96}$, we put $j=0$ in (1)

$$u_{i,1} = \frac{1}{6} [u_{i-1,0} + u_{i+1,0} + 4u_{i,0}]$$

Put $i=1, 2, 3$

$$\begin{aligned} u_{1,1} &= \frac{1}{6} [u_{0,0} + u_{2,0} + 4u_{1,0}] \\ &= \frac{1}{6} [0 + 1 + 4(\frac{1}{\sqrt{2}})] \\ &= 0.6380 \end{aligned}$$

$$\begin{aligned} u_{2,1} &= \frac{1}{6} (u_{1,0} + u_{3,0} + 4u_{2,0}) \\ &= \frac{1}{6} (\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 4(1)) \\ &= 0.9023 \end{aligned}$$

$$\begin{aligned} u_{3,1} &= \frac{1}{6} (u_{2,0} + u_{4,0} + 4u_{3,0}) \\ &= \frac{1}{6} (1 + \frac{4}{\sqrt{2}}) \quad (\because u_{4,0} = 0) \end{aligned}$$

$$u_{3,1} = 0.63807$$

For the second time level $t=2k=\frac{1}{48}$, we put $j=1$ in (1)

$$u_{i,2} = \frac{1}{6} [u_{i-1,1} + u_{i+1,1} + 4u_{i,1}]$$

For $i=1, 2, 3$.

$$\begin{aligned} u_{1,2} &= \frac{1}{6} (u_{0,1} + u_{2,1} + 4u_{1,1}) \\ &= \frac{1}{6} (0 + 0.9023 + 4(0.6380)) \\ &= 0.5757 \end{aligned}$$

4) Solve the Boundary value problem $u_t = u_{xx}$ subject to the conditions $u(0,t) = 0 = u(5,t)$ and $u(x,0) = x^2(25-x^2)$ by taking $h=1$ for $0 \leq t \leq 2$ using Bendre-Schmidt method.

Soln: Given $c^2=1$, $h=1$, $\alpha = \frac{1}{2}$

$$\Rightarrow \alpha = \frac{c^2 k}{h^2} \Rightarrow k = \frac{1}{2}$$

x_0	x_1	x_2	x_3	x_4	x_5
0	1	2	3	4	5

$$u(0,t) = u(x_0, t_j) = u_{0,j} = 0 \quad \text{for } j=0,1,2,3,$$

$$\Rightarrow u_{0,0} = u_{0,1} = u_{0,2} = u_{0,3} = \dots = 0$$

$$u(5,t) = u(x_5, t_j) = u_{5,j} = 0 \quad \text{for } j=0,1,2,3, \dots$$

$$\Rightarrow u_{5,0} = u_{5,1} = u_{5,2} = u_{5,3} = u_{5,4} = \dots = 0$$

Initial condition

$$u(x,0) = x^2(25-x^2)$$

$$u(x_i, 0) = x_i^2(25-x_i^2)$$

$$u_{i,0} = (ih)^2(25-(ih)^2) \quad (h=1)$$

$$u_{i,0} = i^2(25-i^2)$$

$$u_{1,0} = 24, \quad u_{2,0} = 84, \quad u_{3,0} = 144, \quad u_{4,0} = 144$$

By the Bendre Schmidt formula, we have

$$u_{i,j+1} = \frac{u_{i+1,j} + u_{i-1,j}}{2}$$

For the 'first time level we put $j=0$ in ① (i.e $t=0.5$)

$$\begin{aligned}
 u_{2,2} &= \frac{1}{6} (u_{1,1} + u_{3,1} + 4u_{2,1}) \\
 &= \frac{1}{6} (0.6380 + 0.6380 + 4(0.9023)) \\
 &= 0.8142
 \end{aligned}$$

$$\begin{aligned}
 u_{3,2} &= \frac{1}{6} (u_{2,1} + u_{4,1} + 4u_{3,1}) \\
 &= \frac{1}{6} (0.9023 + 0 + 4(0.6380)) \\
 &= 0.5757
 \end{aligned}$$

For the third time level $t=3k=\frac{1}{32}$, we put $j=2$ in (i)

$$u_{i,3} = \frac{1}{6} (u_{i-1,2} + u_{i+1,2} + 4u_{i,2})$$

For $i=1, 2, 3$

$$\begin{aligned}
 u_{1,3} &= \frac{1}{6} (u_{0,2} + u_{2,2} + 4u_{1,2}) \\
 &= \frac{1}{6} (0 + 0.8142 + 4(0.5757)) \\
 &= 0.5195
 \end{aligned}$$

$$\begin{aligned}
 u_{2,3} &= \frac{1}{6} (u_{1,2} + u_{3,2} + 4u_{2,2}) \\
 &= \frac{1}{6} (0.5757 + 0.5757 + 4(0.8142)) \\
 &= 0.7347
 \end{aligned}$$

$$\begin{aligned}
 u_{3,3} &= \frac{1}{6} (u_{2,2} + u_{4,2} + 4u_{3,2}) \\
 &= \frac{1}{6} (0.8142 + 0 + 4(0.5757)) \\
 &= 0.5195
 \end{aligned}$$

$$u_{i,1} = \frac{u_{i+1,0} + u_{i-1,0}}{2} \quad \text{for } i=1,2,3,4$$

$$\Rightarrow u_{1,1} = \frac{u_{2,0} + u_{0,0}}{2} = \frac{84+0}{2} = 42$$

$$u_{2,1} = \frac{u_{3,0} + u_{1,0}}{2} = \frac{144+24}{2} = 84$$

$$u_{3,1} = \frac{u_{4,0} + u_{2,0}}{2} = \frac{144+84}{2} = 114$$

$$u_{4,1} = \frac{u_{5,0} + u_{3,0}}{2} = \frac{0+144}{2} = 72$$

For the second time level, we put $j=1$ in (1), we get
(i.e. $t=1$)

$$u_{i,2} = \frac{u_{i+1,1} + u_{i-1,1}}{2} \quad \text{for } i=1,2,3,4$$

$$u_{1,2} = 42, \quad u_{2,2} = 78, \quad u_{3,2} = 78, \quad u_{4,2} = 57$$

For the third time level, we put $j=2$ in (1), we get
(i.e. $t=1.5$)

$$u_{i,3} = \frac{u_{i+1,2} + u_{i-1,2}}{2} \quad \text{for } i=1,2,3,4$$

$$u_{1,3} = \frac{u_{2,2} + u_{0,2}}{2} = \frac{78+0}{2} = 39$$

$$u_{2,3} = 60, \quad u_{3,3} = 67.5, \quad u_{4,3} = 39$$

For the Fourth time level, we put $j=3$ in (1), we get
(i.e. $t=2$)

$$u_{i,4} = \frac{u_{i+1,3} + u_{i-1,3}}{2} \quad \text{for } i=1,2,3,4$$

$$u_{1,4} = 30, \quad u_{2,4} = 53.25, \quad u_{3,4} = 49.5, \quad u_{4,4} = 33.75$$